

Marking Scheme for HCI 2018 Prelim Paper 2

Qn	Suggested Solution	Marking Scheme
1 [8] (a)	$\int \frac{x+1}{x^2+3x+9} dx$ $= \frac{1}{2} \int \frac{2x+2+1-1}{x^2+3x+9} dx$ $= \frac{1}{2} \int \frac{2x+3}{x^2+3x+9} dx - \frac{1}{2} \int \frac{1}{x^2+3x+9} dx$ $= \frac{1}{2} \ln x^2+3x+9 - \frac{1}{2} \int \frac{1}{\left(x+\frac{3}{2}\right)^2+9-\frac{9}{4}} dx$ $= \frac{1}{2} \ln x^2+3x+9 - \frac{1}{2} \int \frac{1}{\left(x+\frac{3}{2}\right)^2+\left(\frac{\sqrt{27}}{4}\right)^2} dx$ $= \frac{1}{2} \ln x^2+3x+9 - \frac{1}{3\sqrt{3}} \tan^{-1}\left(\frac{2x+3}{3\sqrt{3}}\right) + C$	M1 – Correct separation of terms M1 – Apply $\int \frac{f'(x)}{f(x)} dx$ or Complete the square and apply MF26 formula to obtain $\frac{1}{k} \tan^{-1}\left(\frac{2x+3}{k}\right)$ A1 – Either $\frac{1}{2} \ln x^2+3x+9$ or $\frac{1}{3\sqrt{3}} \tan^{-1}\left(\frac{2x+3}{3\sqrt{3}}\right)$ A1 – Correct answer with +C
1(b)	$\frac{dx}{d\theta} = -\sin \theta, \frac{1}{\sqrt{2}} = \cos \theta \Rightarrow \theta = \frac{\pi}{4}$ $0 = \cos \theta \Rightarrow \theta = \frac{\pi}{2}$ $\int_0^{\frac{1}{\sqrt{2}}} \frac{x^2}{\sqrt{1-x^2}} dx = \int_{\frac{\pi}{2}}^{\frac{\pi}{4}} \frac{\cos^2 \theta}{\sqrt{1-\cos^2 \theta}} \cdot (-\sin \theta) d\theta$ $= \int_{\frac{\pi}{2}}^{\frac{\pi}{4}} (-\cos^2 \theta) d\theta$ $= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos^2 \theta d\theta$ $= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{1+\cos 2\theta}{2} d\theta$ $= \left[-\frac{1}{2}\theta + \frac{1}{4}\sin 2\theta \right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$ $= \frac{\pi}{4} - \left(\frac{\pi}{8} + \frac{1}{4} \right)$ $= \frac{\pi}{8} - \frac{1}{4}$	B1 – Obtain $\theta = \frac{\pi}{4}$ and $\theta = \frac{\pi}{2}$ and differentiate correctly to get $\frac{dx}{d\theta} = -\sin \theta$ M1 – and apply substitution to the integral and obtain $\int_{\theta_1}^{\theta_2} \cos^2 \theta d\theta$ M1 – Attempt to apply double angle formula and able to integration $\cos 2\theta$ correctly A1 – Correct answer in exact form

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2[9] (i)	$\frac{1}{1 + \sin x} = (1 + \sin x)^{-1} = 1 - (\sin x) + (\sin x)^2 - (\sin x)^3 + \dots$ $= 1 - \left(x - \frac{x^3}{3!}\right) + \left(x - \frac{x^3}{3!}\right)^2 - \left(x - \frac{x^3}{3!}\right)^3 + \dots$ $= 1 - x + x^2 - \frac{5}{6}x^3 + \dots$	M1 – Correct binomial expansion $(1 + \sin x)^{-1}$ M1 – Substitute $\sin x \approx x - \frac{x^3}{3!}$ A1 – Correct answer
2(ii)	Replace x by x^3, $\frac{1}{1 + \sin x^3} = 1 - x^3 + x^6 - \frac{5}{6}x^9 + \dots$ $\Rightarrow \frac{f^9(0)}{9!} = -\frac{5}{6}$ $\Rightarrow f^9(0) = -\frac{5}{6} \times 9! = -302400$	M1 – Replace x by x^3 A1 – Correct answer
2(iii)	$\left(1 + \frac{x}{a}\right)^b = 1 + b\left(\frac{x}{a}\right) + \frac{b(b-1)}{2}\left(\frac{x}{a}\right)^2 + \frac{b(b-1)(b-2)}{6}\left(\frac{x}{a}\right)^3 + \dots$ $\frac{1}{1 - \sin x} = 1 + x + x^2 + \frac{5}{6}x^3 + \dots$ Coefficient of x: $\frac{b}{a} = 1 \Rightarrow a = b \dots (1)$ Coefficient of x^2: $\frac{b(b-1)}{2a^2} = 1 \dots (2)$ Sub (1) into (2): $\frac{b(b-1)}{2b^2} = 1$ $\Rightarrow \frac{(b-1)}{2b} = 1$ $\Rightarrow b-1 = 2b$ $\Rightarrow b = 1 \dots (3)$ Coefficient of x^3: $\frac{b(b-1)(b-2)}{6a^3} = -\frac{5}{6} \dots (4)$ Sub (1) and (2) into (4): $\frac{b(b-1)(b-2)}{6a^3} = \frac{5}{6}$ $\Rightarrow \frac{b(b-1)}{2a^2} \cdot \frac{b-2}{3a} = \frac{5}{6}$ $\Rightarrow \frac{b-2}{3b} = \frac{5}{6}$ $\Rightarrow b = \frac{4}{3} \dots (5)$	B1 – Correct expansion for $\left(1 + \frac{x}{a}\right)^b$ B1 – Correct expansion for $\frac{1}{1 - \sin x}$ M1 – or at least 2 correct equation out of 3 (eqs 1,2,4) And attempt for attempt to solve simult equations involving a and b A1 – Correct conclusion

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	Results in (3) and (4) contradict each other. Hence the statement is invalid.	
3[11] (i)	Since the line $y = a$ cuts the graph $y = f(x)$ more than once, f is not a one-one function and therefore f^{-1} does not exist.	G1 – Correct shape of graph for $y = ae^{a-x}$ for $0 \leq x < a$. G1 – Correct shape of graph for $y = \left \frac{1}{a}(x-a)^2 - a \right$ for $x \geq a$ G1 – Axial intercepts B1 – Give a counter example to show inverse does not exist
3(ii)	Largest value of $k = 2a$	B1
3(iii)	Let $y = f(x) \Rightarrow f^{-1}(y) = x$ 1. $y = a - \frac{1}{a}(x-a)^2$ $(x-a)^2 = a^2 - ay$ $x = a \pm \sqrt{a^2 - ay}$ (reject $x = a - \sqrt{a^2 - ay}$ $\because x \geq a$) $\therefore x = a + \sqrt{a^2 - ay}$ 2. $y = ae^{a-x}$ $a - x = \ln \frac{y}{a}$ $x = a - \ln \frac{y}{a}$ $\therefore f^{-1} : x \mapsto \begin{cases} a + \sqrt{a^2 - ax} & \text{for } 0 \leq x \leq a \\ a - \ln \frac{x}{a} & \text{for } a < x \leq ae^a \end{cases}$	M1 – attempt to make x the subject for both expressions of $f(x)$ M1 – at least one of the expressions of $f^{-1}(y)$ correct A1 – correct answer with domain, in similar form Do not award full mark if student did not explain

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3(iv)	$f\left(\frac{a}{2}\right) = ae^{\frac{a}{2}} = ae^{\frac{a}{2}}$ $ff\left(\frac{a}{2}\right) = f\left[f\left(\frac{a}{2}\right)\right]$ $= f\left(ae^{\frac{a}{2}}\right)$ $= a - \frac{1}{a}\left(ae^{\frac{a}{2}} - a\right)^2$ $= a - a\left(e^{\frac{a}{2}} - 1\right)^2$ $= ae^{\frac{a}{2}}\left(2 - e^{\frac{a}{2}}\right)$ Since $0 < a \leq \ln 2$ $0 < \frac{a}{2} \leq \ln \sqrt{2}$ $1 < e^{\frac{a}{2}} \leq \sqrt{2}$ $a < ae^{\frac{a}{2}} \leq a\sqrt{2} < 2a$ Alternatively, $f(x) = \begin{cases} ae^{a-x} & \text{for } 0 \leq x < a \\ a - \frac{1}{a}(x-a)^2 & \text{for } a \leq x \leq 2a \end{cases}$ $R_f = \begin{cases} (a, ae^a] \\ [0, a] \end{cases}$ $\therefore ff(x) = \begin{cases} a - \frac{1}{a}(ae^a - a)^2 = a - a(e^a - 1)^2 & \text{for } 0 \leq x < a \\ ae^{a-a+\frac{1}{a}(x-a)^2} = ae^{\frac{1}{a}(x-a)^2} & \text{for } a \leq x \leq 2a \end{cases}$ $ff\left(\frac{a}{2}\right) = a - a\left(e^{\frac{a}{2}} - 1\right)^2 = ae^{\frac{a}{2}}\left(2 - e^{\frac{a}{2}}\right)$	M1 – attempt to substitute $ae^{\frac{a}{2}}$ using the correct expression A1 – correct answer OR M1 – finding $ff(x)$ with at least one of the expressions correct A1 – correct answer

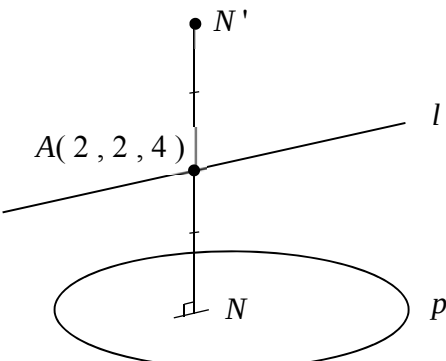
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	Note for students: for ff to exist, $R_f \subseteq D_f$ $R_f = [0, ae^a]$ $D_f = [0, 2a]$ $\Rightarrow ae^a \leq 2a$ $a(e^a - 2) \leq 0$ Since $a > 0$, $e^a - 2 \leq 0$ $a \leq \ln 2$ $\therefore 0 < a \leq \ln 2$	
4[12] (i)	Let $t = x - 1 = \frac{y}{2} = z - 3$, $t \in \mathbb{R}$. $\therefore x = 1 + t$ $y = 0 + 2t$ $z = 3 + t$ Hence $l: r = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} + t \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$, $t \in \mathbb{R}$. Normal of p is $\begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} \times \begin{pmatrix} k \\ 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 7 \\ -8-3k \\ 2-k \end{pmatrix}$ Since l and p are parallel, $\begin{pmatrix} 7 \\ -8-3k \\ 2-k \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$ are perpendicular. Hence $\begin{pmatrix} 7 \\ -8-3k \\ 2-k \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = 0$ $7 - 16 - 6k + 2 - k = 0$ $\therefore k = -1 \quad (\text{shown})$	B1: obtain correct direction vector of l M1: apply cross product on 2 direction vectors AG1: apply dot product $\begin{pmatrix} 7 \\ -8-3k \\ 2-k \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = 0 \text{ and obtain } k = -1$
4(ii)	<u>Method 1:</u> (using equation of p in dot product form) $p: r \cdot \begin{pmatrix} 7 \\ -5 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ -2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 7 \\ -5 \\ 3 \end{pmatrix} = 48$	M1: obtain $p: r \cdot \begin{pmatrix} 7 \\ -5 \\ 3 \end{pmatrix} = 48$

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	Substitute a point $(1,0,3)$ on l into p, $\begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 7 \\ -5 \\ 3 \end{pmatrix} = 16 \neq 48$ Hence l does not lie on p. $\therefore l$ and p do not intersect. (shown) <u>Method 2:</u> (using equation of p in parametric form) If l and p intersect, $\begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} + t \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ -2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 1 \\ 4 \end{pmatrix}$ $\begin{pmatrix} 1+t \\ 2t \\ 3+t \end{pmatrix} = \begin{pmatrix} 5+2\lambda-\mu \\ -2+\lambda+\mu \\ 1-3\lambda+4\mu \end{pmatrix}$ $\begin{aligned} \therefore \quad 2\lambda - \mu - t &= -4 \\ \lambda + \mu - 2t &= 2 \\ -3\lambda + 4\mu - t &= 2 \end{aligned}$ Solving using GC, no solution. $\therefore l$ and p do not intersect. (shown)	A1: show l does not lie on p and conclude l and p do not intersect. M1: equating l and p A1: show system of equations has no solution and conclude l and p do not intersect.
4(iii)	<u>Method 1:</u> (using $\left \frac{a \cdot b}{ a b } \right$) Using points $(1,0,3)$ on l and $(5,-2,1)$ on p, $\begin{pmatrix} 5 \\ -2 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ -2 \\ -2 \end{pmatrix}$ Hence required distance = $\left \begin{pmatrix} 4 \\ -2 \\ -2 \end{pmatrix} \cdot \frac{1}{\sqrt{83}} \begin{pmatrix} 7 \\ -5 \\ 3 \end{pmatrix} \right$ $= \frac{32}{\sqrt{83}} = \frac{32\sqrt{83}}{83} \text{ units}$ <u>Method 2:</u> (using intersection of $\perp$ line with p) Using the point $(1,0,3)$ on l, Equation of line through $(1,0,3)$ and perpendicular to p is	M1: attempt to find a vector from l to p M1: use $\left \frac{a \cdot b}{ a b } \right$ to find distance A1: correct answer

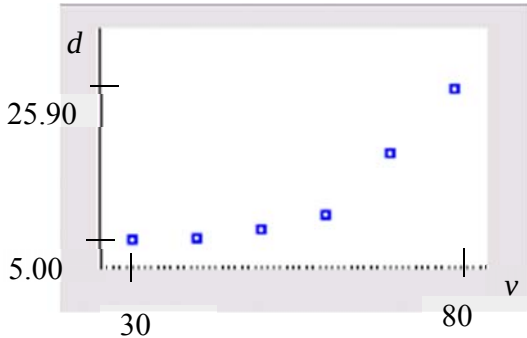
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	$r = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} + s \begin{pmatrix} 7 \\ -5 \\ 3 \end{pmatrix}, \quad s \in \mathbb{R}$ When line intersects p, $\begin{pmatrix} 1+7s \\ -5s \\ 3+3s \end{pmatrix} \cdot \begin{pmatrix} 7 \\ -5 \\ 3 \end{pmatrix} = 48$ $7 + 49s + 25s + 9 + 9s = 48$ $\therefore s = \frac{32}{83}$ Hence required distance = $\left \frac{32}{83} \begin{pmatrix} 7 \\ -5 \\ 3 \end{pmatrix} \right = \frac{32\sqrt{83}}{83}$ units	M1: substitute equation of line into equation of p to find intersection M1: use $\left s \begin{pmatrix} 7 \\ -5 \\ 3 \end{pmatrix} \right$ to find required distance A1: correct answer
4(iv)	 <u>Method 1:</u> (using part (iii)) From (iii), $\vec{AN} = \frac{32\sqrt{83}}{83} \left[\frac{1}{\sqrt{83}} \begin{pmatrix} 7 \\ -5 \\ 3 \end{pmatrix} \right] = \frac{32}{83} \begin{pmatrix} 7 \\ -5 \\ 3 \end{pmatrix}$ $\therefore \vec{ON} = \vec{OA} + \vec{AN} = \begin{pmatrix} 2 \\ 2 \\ 4 \end{pmatrix} + \frac{32}{83} \begin{pmatrix} 7 \\ -5 \\ 3 \end{pmatrix} = \frac{1}{83} \begin{pmatrix} 390 \\ 6 \\ 428 \end{pmatrix}$ Hence $N\left(\frac{390}{83}, \frac{6}{83}, \frac{428}{83}\right)$ [or $N(4.70, 0.0723, 5.16)$] <u>Method 2:</u> (using intersection of $\perp$ line through A with p) Equation of line through $(2, 2, 4)$ and perpendicular to p is	M1: attempt to find $\vec{AN}$ using $\left[\begin{pmatrix} 4 \\ -2 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} 7 \\ -5 \\ 3 \end{pmatrix} \right]$ A1: correct coordinates of N.

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	$r_{\%} = \begin{pmatrix} 2 \\ 2 \\ 4 \end{pmatrix} + \gamma \begin{pmatrix} 7 \\ -5 \\ 3 \end{pmatrix}, \quad \gamma \in \mathbb{R}$ When line intersects p, $\begin{pmatrix} 2+7\gamma \\ 2-5\gamma \\ 4+3\gamma \end{pmatrix} \cdot \begin{pmatrix} 7 \\ -5 \\ 3 \end{pmatrix} = 48$ $14 + 49\gamma - 10 + 25\gamma + 12 + 9\gamma = 48$ $\therefore \gamma = \frac{32}{83}$ $\therefore \vec{ON} = \begin{pmatrix} 2 \\ 2 \\ 4 \end{pmatrix} + \frac{32}{83} \begin{pmatrix} 7 \\ -5 \\ 3 \end{pmatrix} = \frac{1}{83} \begin{pmatrix} 390 \\ 6 \\ 428 \end{pmatrix}$ Hence $N\left(\frac{390}{83}, \frac{6}{83}, \frac{428}{83}\right)$ [or $N(4.70, 0.0723, 5.16)$ 3 s.f.] Let $N'(x, y, z)$ be the reflection of N in l. $\therefore \begin{pmatrix} 2 \\ 2 \\ 4 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} x + \frac{390}{83} \\ y + \frac{6}{83} \\ z + \frac{428}{83} \end{pmatrix}$ $x = -\frac{58}{83}, \quad y = \frac{326}{83}, \quad z = \frac{236}{83}$ $\therefore N'\left(-\frac{58}{83}, \frac{326}{83}, \frac{236}{83}\right)$ [or $N'(-0.699, 3.93, 2.84)$]	M1: substitute equation of line into equation of p to find intersection A1: correct coordinates of N. M1: use mid-point theorem to find N' A1: correct coordinates of N'
5(i)	$P(Y \text{ is odd}) = \frac{\frac{1}{4}}{1 - \frac{1}{4}} = \frac{1}{3}$	M1 – Using correct GP sum formula with either a or r correct. A1 – correct answer.

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5(ii)	$P(Y_1 + Y_2 < 4 Y_1 + Y_2 > 2)$ $= \frac{P(2 < Y_1 + Y_2 < 4)}{P(Y_1 + Y_2 > 2)} = \frac{P(Y_1 + Y_2 = 3)}{1 - P(Y_1 + Y_2 \leq 2)}$ $= \frac{P((Y_1, Y_2) = (0, 3), (1, 2), (2, 1), (3, 0))}{1 - P((Y_1, Y_2) = (0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (2, 0))}$ $= \frac{2 \left[\frac{1}{2} \frac{1}{16} + \frac{1}{4} \frac{1}{8} \right]}{1 - \left[\frac{1}{2} \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{8} \right) + \frac{1}{4} \left(\frac{1}{2} + \frac{1}{4} \right) + \frac{1}{8} \frac{1}{2} \right]} = \frac{\frac{1}{8}}{\frac{5}{16}} = \frac{2}{5}$	M1 – Getting $\frac{P(Y_1 + Y_2 = 3)}{P(Y_1 + Y_2 > 2)}$. M1 – Getting 6 cases for denominator in the complement method, or 10 cases for denominator in the direct method. A1 – Getting numerator or denominator correctly. A1 – Correct answer.
6(i)	No of ways = $2^6 6! = 46080$	M1 – Getting 2^6 or $6!$ A1 – Correct answer.
6(ii)	<u>Method 1</u> No of ways = $6! \binom{5}{C_1} 5! (2) = 864000$ <u>Method 2</u> No of ways with all women next to one another and all men next to one another = $6! 6! 2 = 1036800$ No of ways with couple seated at 6th and 7th seat together, and all women next to one another and all men next to one another = $5! 5! (6)(2) = 172800$ Required no of ways = $1036800 - 172800 = 864000$	M1 – Getting $6! 5!$ or $5! 5!$ M1 – Multiply by 2. A1 – Correct answer. M1 – Complement method with either $6! 6!$ or $5! 5! 6$ obtained. M1 – Multiply by 2. A1 – Correct answer.
	No of arrangement = $(5-1)! 2 = 48$ Probability required = $\frac{48}{9!} = \frac{1}{7560}$ (or 0.000132 (3 s.f.))	M1 – Getting $(5-1)! = 24$ A1 – Correct answer.
7(i)		B1 – Correct relative positions of points for scatter plot B1 – Correct labelling of axes.
7(ii) (a)	0.902 (3 s.f.)	B1 – Correct r value

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7(ii) (b)	0.955 (3 s.f.)	B1 – Correct r value
7(iii)	From the scatter plot of the data, it seems to fulfil the traits of exponential curve rather than a linear curve. For Model II, the r value is closer to 1 as compared to Model I. Thus Model II is better.	B1 – Correct explanation
7(iv)	$d = e^{cv+d}$ $\ln d = cv + d$ Regression line of $\ln d$ on v : $\ln d = 0.0342758025v + 0.34294554177$ $\ln d = 0.0343v + 0.343$ (3 s.f.)	B1 – Correct regression line of $\ln d$ on v .
7(v)	When $d = 10$, $\ln 10 = 0.0342758025v + 0.34294554177$ $v = 57.2$ km/h	B1 – Correct answer
7(vi)	$d = 0.4256v - 11.74$ $(\bar{d}, \bar{v})$ satisfies the regression line. $\Rightarrow \bar{d} = 0.4256\bar{v} - 11.74$ Let d be the distance travelled after brakes are applied. $\Rightarrow \frac{(68.15 + d)}{7} = 0.4256 \frac{(330 + 75)}{7} - 11.74$ $\Rightarrow 68.15 + d = (0.4256)(405) - (11.74)(7)$ $\Rightarrow d = 22.04$ metres	M1 – using $(\bar{d}, \bar{v})$ on regression line. M1 – calculating $\bar{d}$ or $\bar{v}$ correctly. A1 – correct answer
8(i)	Let X denote the time taken to finish a meal. $\bar{x} = \frac{1}{n} \sum x = \frac{1380}{50} = 27.6$ $s^2 = \frac{1}{n-1} \left(\sum x^2 - \frac{(\sum x)^2}{n} \right)$ $= \frac{1}{49} \left(83000 - \frac{1380^2}{50} \right)$ $= \frac{44912}{49} = 916.57$ $= 917$ (3 s.f.)	B1 – correct unbiased estimate of population mean B1 – correct unbiased estimate of population variance
8(ii)	Let μ be the population mean time taken for a student to finish a meal during recess.	B1 – Correct H_0 and H_1

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	$H_0 : \mu = 20$ $H_1 : \mu > 20$ Under H_0 , test statistic, $Z = \frac{\bar{X} - 20}{\frac{\sqrt{44912/49}}{\sqrt{50}}} \sim N(0, 1)$ approximately by Central Limit Theorem, since $n = 50$ is sufficiently large. p -value = 0.0379 (3 s.f.) Since p -value = 0.0379 < 0.05, we reject H_0 at 5% level of significance. There is sufficient evidence to conclude that the time taken by students is more than 20 minutes. Assumption: Time taken by each student is independent of each other and that the students are chosen randomly.	M1 – Doing Z-test with correct standardisation A1 – Correct p -value A1 – Correct conclusion (award only if p -value is correct) B1 – Correct assumption
8(iii)	5% level of significance means there is a probability of 0.05 of concluding that the population mean time taken by a student to finish a meal during recess is more than 20 minutes when in fact, the <u>population mean time is not more than 20 minutes</u> .	B1 – Correct definition in context. (accept if underlined phrase is “population mean time is 20 minutes”)
8(iv)	Jack did the Z-test with 1-tail test and rejected H_0 . Jill did the Z-test with 2-tail test. p -value for 1-tail test = 0.0379, which is < 0.05. With the same test statistic, under Z-test 2-tail test, the p -value will be doubled, i.e. $0.0379 \times 2 = 0.0758 > 0.05$. Jill will not be rejecting H_0 and have a different conclusion from Jack.	M1 – p -value in 2 tail test = $2(p$ -value in 1 tail test) A1 – Correct conclusion
9(i)	Binomial distribution. The colour of a ball chosen is independent of another.	B1 – Correct reason
9(ii)	Let Y denote no of red balls obtained in $2n$ draws from the bag. $Y : B\left(2n, \frac{1}{3}\right)$ $P(Y \leq 10) \leq 0.5$ When $n = 15$, $P(Y \leq 10) = 0.5848 > 0.5$ When $n = 16$, $P(Y \leq 10) = 0.4836 < 0.5$ Thus least $n = 16$.	M1 – Distribution of Y and inequality $P(Y \leq 10) \leq 0.5$ M1 – giving probabilities for 2 cases $n=15$ and $n=16$. A1 – Correct answer.
9(iii)	Let S denote no of red balls obtained by Sue. $S : B\left(5, \frac{1}{3}\right)$ Let B denote no of red balls obtained by Ben. $B : B\left(5, \frac{1}{3}\right)$ Required probability	

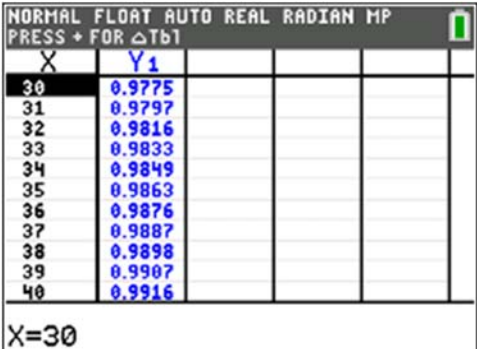
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	$= P(\text{Sue wins} \mid \text{gets} > 3 \text{ red balls})$ $= \frac{P(\text{Sue wins and gets} > 3 \text{ red balls})}{P(\text{gets} > 3 \text{ red balls})}$ $= \frac{P(S=4)P(B \leq 3) + P(S=5)P(B \leq 4)}{1 - P(S \leq 3)}$ $= \frac{0.0433877}{0.045267}$ $= 0.958$	M1 – apply conditional probability M1 – correct numerator cases A1 – correct answer (3 s.f.)
9(iv)	Let X denote number of balls drawn before Sue obtains 2 yellow balls. $P(X=2) = P(YY) = \frac{3}{6} \cdot \frac{2}{5} = \frac{1}{5}$ $P(X=3) = P(YY'Y, Y'YY) = \frac{3}{6} \cdot \frac{3}{5} \cdot \frac{2}{4} \cdot 2 = \frac{3}{10}$ $P(X=4) = P(YY'Y'Y, Y'YY'Y, Y'Y'YY) = \frac{3}{6} \cdot \frac{3}{5} \cdot \frac{2}{4} \cdot \frac{2}{3} \cdot 3 = \frac{3}{10}$ $P(X=5)$ $= P(YY'Y'Y'Y, Y'YY'Y'Y, Y'Y'YY'Y, Y'Y'Y'YY)$ $= \frac{3}{6} \cdot \frac{3}{5} \cdot \frac{2}{4} \cdot \frac{1}{3} \cdot \frac{2}{2} \cdot 4 = \frac{1}{5}$ Expected number $= (2)\left(\frac{1}{5}\right) + (3)\left(\frac{3}{10}\right) + (4)\left(\frac{3}{10}\right) + (5)\left(\frac{1}{5}\right)$ $= 3.5$ (or by observation of data in table)  Use $\text{Var}(X) = E(X^2) - [E(X)]^2$	B1 – correct probability for 2 out of the 4 B1 – All probabilities correct B1 – Correct Expectation M1 – Correct method to get variance (SOI) A1 – Var (X) correct

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	$\text{Var}(X) = 1.05 = \frac{21}{20}$	
10	Let D be the time taken to drive to office. $D \sim N(14, 2.1^2)$ Let T be the time held up at a traffic light junction. $T \sim N(\mu, 0.2^2)$ $D + T_1 + T_2 + T_3 + T_4 \sim N(14 + 4\mu, 4.57)$ $P(D + T_1 + T_2 + T_3 + T_4 \leq 20) = 0.713$ $P(Z \leq \frac{20 - (14 + 4\mu)}{\sqrt{4.57}}) = 0.713$ $\frac{6 - 4\mu}{\sqrt{4.57}} = 0.5621702875$ $\mu = 1.19955$ $= 1.2 \text{ (1 d.p.)}$ The driving time and the time held up at the traffic light junctions are independent of one another.	M1 – getting $N(14 + 4\mu, 4.57)$ correctly and formulating $P(D + T_1 + T_2 + T_3 + T_4 \leq 20) = 0.713$ M1 – attempt to solve by standardization or from GC. AG1 – correct answer B1 – correct assumption
10(i)	Probability Tony is late on 2 days in the first 9 days $= {}^9C_2 (0.713)^7 (1 - 0.713)^2 = 0.2777749021$ P(Tony is late at his office for the third time on the 10th day) $= {}^9C_2 (0.713)^7 (1 - 0.713)^2 (1 - 0.713)$ $= 0.0797 \text{ (3 s.f.)}$	M1 – showing $\alpha (0.713)^7 (1 - 0.713)^2$ or showing the use of $B(9, 1 - 0.713)$ A1 – correct answer
10(ii)	$D - 10T \sim N(2, 8.41)$ $P(D < 10T) = P(D - 10T < 0) = 0.245 \text{ (3 s.f.)}$	M1 – getting distn of $D - 10T$ A1 – correct answer
10(iii)	$D - (T_1 + T_2 + T_3 + T_4) \sim N(9.2, 4.57)$ $P(D - (T_1 + T_2 + T_3 + T_4) > 8)$ $= 1 - P(-8 < D - (T_1 + T_2 + T_3 + T_4) < 8)$ $= 1 - 0.28728$ $= 0.713 \text{ (3 s.f.)}$	M1 – getting $N(9.2, 4.57)$ with at least one of 2 parameters correct M1 – formulate correctly $P(D - (T_1 + T_2 + T_3 + T_4) > 8)$ and attempt to remove modulus A1 – correct answer
10(iv)	Let μ be the population mean time held up at a traffic light junction, $\bar{X}$ be the mean time held up at a traffic light in a sample of size n. $\bar{X} \sim N\left(\mu, \frac{0.2^2}{n}\right)$	M1 – getting distn of $\bar{X}$ correctly and formulating $P\left(\bar{X} - \mu \leq \frac{1}{12}\right) \geq 0.98$

Marking Scheme for HCI 2018 Prelim Paper 2

Qn	Suggested Solution	Marking Scheme
	Given $P\left(\left \bar{X} - \mu\right \leq \frac{1}{12}\right) \geq 0.98$ $P\left(\left Z\right \leq \frac{\frac{1}{12}}{\frac{0.2}{\sqrt{n}}}\right) \geq 0.98$ $P\left(-\frac{5\sqrt{n}}{12} \leq Z \leq \frac{5\sqrt{n}}{12}\right) \geq 0.98$ $\frac{5\sqrt{n}}{12} \geq 2.326347877$ $n \geq 31.17$ Smallest $n = 32$ OR Using GC Tables, $n = 31, P\left(-\frac{5\sqrt{n}}{12} \leq Z \leq \frac{5\sqrt{n}}{12}\right) = 0.9797$ $n = 32, P\left(-\frac{5\sqrt{n}}{12} \leq Z \leq \frac{5\sqrt{n}}{12}\right) = 0.9816$ 	M1 – attempt to solve by standardizing or showing list of GC table values A1 – correct answer